

Giona Sala

📍 Nationality: Swiss

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in gionasala

Research experience

- PhD Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen,** Aachen, DE
 Doctorate in cosmology and gravitational wave physics
 Oct 2023 – present
- Topic: *The Rise of Gravitational Waves from Compact Binaries: Observational Prospects and Cosmological Implications*
 - Supervisors: Dr. Sebastian Baum & Prof. Julien Lesgourgues

Education

- MSc École Polytechnique Fédérale de Lausanne (EPFL),** Lausanne, CH
 Master of Science in Physics
 Sep 2020 – Apr 2023
- Focus on Theoretical Particle Physics
 - Minor in space technologies
- Université de Genève (UniGe),** Geneva, CH
 Master's thesis
 Sep 2022 – Apr 2023
- Title: *Implications of Primordial Black Holes physics at high redshifts*
 - Supervisor: Prof. Antonio Riotto
- BSc Eidgenössische Technische Hochschule Zürich (ETHZ),** Zürich, CH
 Bachelor of Science in Physics
 Sep 2017 – Sep 2020
- Final project on Hardy's Paradox
- University of Toronto (UofT),** Toronto, CA
 Exchange
 Sep 2019 – Dec 2019

Publications

in G.Sala.5

0009-0001-3716-862X

SalaPh

Primary contribution

- Detecting White Dwarf Binary Mergers with Gravitational Waves** Jun 2026
Giona Sala, Chiara Brandenstein, Sebastian Baum, Peter W. Graham
 Phys. Rev. D 113 (2026) 123022 [↗](#)
- Inferring cosmological parameters from galaxy and dark sirens cross-correlation** May 2026
Giona Sala, Alessandro Cuoco, Julien Lesgourgues, Kostantinos-Rafail Revis, Lorenzo Valbusa Dall'Armi, Santiago Casas
 JCAP 05 (2026) 095 [↗](#)

Supporting contribution

- Cosmology beyond standard sirens: cross-correlation of gravitational waves and neutral hydrogen intensity mapping** July 2026
 Matteo Schulz, Ulyana Dupletsa, Andrea Cozzumbo, **Giona Sala**, Tommaso Ronconi, Marta Spinelli, Riccardo Murgia, Simone Mastrogiovanni, Jan Harms, Erik de la Haye, Felicitas Keil
 arXiv:2607.26193 [astro-ph.CO] [↗](#)

Conference contributions & talks

Swiss Cosmology Days 2026

Geneva, CH
Jun 2026

- Presentation on *Pushing cosmological inference with next-generation gravitational wave surveys*

3rd Sardinian Gravitational Wave Science (SGWS) International Mini-Workshop

Cagliari, IT
May 2026

- Presentation on *Can we detect the anisotropies of the stochastic gravitational waves background?*

Gran Sasso Science Institute (GSSI) seminar

L'Aquila, IT
Feb 2025

- Invited talk on *The Hubble tension in the GW era: A Cross-Correlation study*

Seminar on the Interface of Particle Physics and Gravitational Waves

Online
Jan 2025

- Presentation on *Detecting White Dwarf Binary Mergers with Gravitational Waves*

MAGIS-100 meetings

Online (Fermilab, US)
Nov 2025

- Presentation on *Detecting White Dwarf Binary Mergers with Gravitational Waves*

Invisibles25 Workshop

CERN, CH
Sep 2025

- Presentation and poster on *Inferring cosmological parameters from galaxy and dark sirens cross-correlation*

WE-Heraeus-Seminar: new windows on the Universe

Kitzbühel, A
May 2025

- Poster on *Inferring cosmological parameters from galaxy and dark sirens cross-correlation*

European Consortium for Astroparticle Theory (EuCAPT) Annual Symposium

Online (CERN, CH)
May 2025

- Plenary lightning talk on *Inferring cosmological parameters from galaxy and dark sirens cross-correlation*

Grants and research visits

CosmoVerse Short-term Scientific Missions (STSM) Grant

Geneva, CH
Sep 2026

- Visit to the research groups of Prof. Camille Bonvin and Prof. Ruth Durrer at the University of Geneva working on *Astrophysical Gravitational Wave Signatures in Large-Scale Structure as Cosmological Probes*.

Additional experiences

Teaching experience

RWTH,

Teaching assistant

Aachen, DE
Apr 2024 – present

- Supervision:
 - Master's thesis on *Component separation algorithms for the gravitational wave backgrounds*
 - Bachelor's thesis on *Separation of the cosmological and astrophysical gravitational wave background*
- Courses: Theory of Relativity and Cosmology x2; The ingredients of the universe; Theoretical Physics I: Mechanics

EPFL,

Teaching assistant

Lausanne, CH
Sep 2021 – Jan 2023

- Courses: General Physics: Electromagnetism x2, General Physics II, Experimental Physics I, Physics IV

Work experience

Fondazione Istituto Ricerche Solari (IRSOL),
Civil Service

Locarno-Monti, CH
Apr 2023 – Aug 2023

- Solar research technical and scientific collaborator

Scuola Universitaria Professionale della Svizzera Italiana (SUPSI),
Civil Service

Mendrisio, CH
Jun 2022 – Jul 2022

- Renewable energy research collaborator

Extracurricular activities

EPFL Spacecraft Team (EST),
System engineer team leader

Lausanne, CH
Sep 2021 – Jun 2022

- Student association that coordinates the CHESS (Constellation of High Energy Swiss Satellites) space mission

EPFL Rocket Team (ERT),
Payload team leader

Lausanne, CH
Sep 2020 – Oct 2021

- Team of students from EPFL building Bella Lui II 2021 rocket for international competitions

Akademische Raumfahrt Initiative Schweiz (ARIS),
Structure engineer support

Zürich, CH
Feb 2020 – Jul 2020

- The student association for space in the german part of Switzerland, buildin EULER 2020 rocket

Skills

Programming languages

Python ●●●●●
LaTeX ●●●●●

C/C++ ●●●●●
MATLAB ●●●●●

Programmes

Mathematica ●●●●●
Excel ●●●●●

MontePython ●●●●●
CLASS ●●●●●

Languages

Italian Native
English ●●●●●

French ●●●●●
German ●●●●●

References

Julien Lesgourgues

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Institute for Theoretical Particle Physics
and Cosmology
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Sebastian Baum

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Mid Sweden University
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Peter W. Graham

Prof. Dr.
Leinweber Institute for Theoretical Physics
Stanford University
✉ pwgraham@stanford.edu

Antonio Riotto

Prof. Dr.
Department of theoretical physics
Université de Genève
✉ antonio.riotto@unige.ch